

WADDINGTON, United Kingdom

WMO: 033770

Lat: 53.1753N

Lon: 0.5233W

Elev: 70

StdP: 100.48

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.8	-1.6	-5.9	2.3	-0.1	-4.7	2.6	0.4	14.4	9.1	13.2	8.8	3.5	330	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	26.1	18.0	24.0	17.1	22.3	16.4	19.0	23.9	18.1	22.4	17.2	21.2	4.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.4	20.7	16.4	11.7	19.7	15.6	11.2	18.9	54.2	24.0	51.2	22.4	48.6	21.3	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.0	10.6	9.4	DB	-5.3	29.8	2.0	1.6	-6.8	31.0	-8.0	31.9	-9.1	32.9	-10.6	34.1	
			WB	-5.8	20.9	1.9	1.2	-7.2	21.7	-8.4	22.4	-9.5	23.1	-10.9	24.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	4.3	4.8	6.4	8.8	11.7	14.6	17.0	16.9	14.3	11.0	7.1	4.7
	DBStd	5.38	3.01	3.18	3.00	2.92	2.90	2.60	2.55	2.41	2.31	2.75	2.94	3.46
	HDD10.0	793	177	148	115	57	15	1	0	0	1	21	94	167
	HDD18.3	3019	434	379	368	287	206	114	58	59	124	228	337	423
	CDD10.0	853	1	2	5	20	67	139	217	213	129	51	7	2
	CDD18.3	37	0	0	0	0	0	4	17	14	2	1	0	0
	CDH23.3	247	0	0	0	1	3	36	112	83	12	2	0	0
	CDH26.7	38	0	0	0	0	0	5	17	16	1	0	0	0
Wind	WSAvg	4.7	5.3	5.3	5.1	4.7	4.6	4.1	4.0	4.1	4.3	4.9	4.9	5.1
Precipitation	PrecAvg	602	50	37	42	46	48	54	55	62	49	54	55	52
	PrecMax	823	116	133	109	155	118	201	133	146	136	132	120	163
	PrecMin	384	12	3	6	7	6	4	12	6	8	9	19	15
	PrecStd	97	24	22	24	30	27	39	32	29	30	30	24	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	13.8	16.0	20.8	23.5	27.3	28.8	28.7	25.1	20.2	15.6	13.6
		MCWB	10.9	10.9	11.0	13.6	16.4	19.1	19.2	18.6	16.9	16.0	13.6	11.6
	2%	DB	11.3	12.0	14.0	17.5	21.1	23.7	26.1	25.8	22.1	17.7	13.9	12.4
		MCWB	9.6	9.7	10.1	12.0	15.0	16.8	18.1	18.0	16.5	14.4	12.2	10.8
	5%	DB	10.1	10.7	12.6	15.5	19.0	21.5	24.2	23.2	20.0	16.3	12.6	11.1
		MCWB	8.6	8.6	9.5	10.9	13.7	15.7	17.3	17.0	15.4	13.8	11.0	9.7
	10%	DB	8.9	9.5	11.3	13.8	17.1	19.7	22.3	21.5	18.5	15.1	11.4	9.8
		MCWB	7.5	7.7	8.5	10.0	12.6	14.8	16.5	16.1	14.4	12.9	10.0	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.5	12.2	14.4	17.3	20.0	20.4	20.1	18.6	16.5	14.0	12.0
		MCDB	12.6	13.3	14.6	19.1	22.2	25.7	26.6	25.2	22.5	19.5	15.3	13.1
	2%	WB	9.9	10.0	11.0	12.8	15.5	17.7	19.1	18.8	17.2	15.3	12.5	11.0
		MCDB	11.0	11.7	13.2	16.4	20.0	21.9	24.1	23.4	20.7	17.2	13.6	12.3
	5%	WB	8.8	9.0	9.9	11.6	14.3	16.5	18.1	17.9	16.1	14.2	11.3	9.8
		MCDB	9.9	10.5	12.2	14.8	18.0	20.2	22.5	22.2	18.8	15.9	12.3	11.0
	10%	WB	7.7	7.9	8.9	10.4	13.2	15.5	17.2	17.0	15.2	13.3	10.2	8.6
		MCDB	8.8	9.3	11.0	13.4	16.3	19.0	21.4	20.5	17.6	14.8	11.2	9.7
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.4	6.5	7.7	8.1	8.1	8.5	8.2	7.4	6.2	5.1	4.5
		MCDBR	6.0	6.3	8.2	10.2	11.0	11.3	11.9	10.7	9.7	7.1	5.8	5.7
		MCWBR	5.5	4.9	5.3	5.8	6.1	5.7	5.2	4.6	5.1	4.7	5.0	5.2
	5% WB	MCDBR	6.0	6.1	7.6	9.6	10.2	10.1	10.5	9.6	8.4	6.9	5.7	5.6
		MCWBR	5.6	5.1	5.5	5.9	6.1	5.8	5.3	4.6	5.2	5.2	5.1	5.3
Clear-Sky Solar Irradiance	taub	0.322	0.338	0.370	0.396	0.397	0.403	0.404	0.391	0.370	0.350	0.324	0.310	
	taud	2.359	2.391	2.298	2.236	2.274	2.281	2.288	2.347	2.411	2.446	2.410	2.334	
	Ebn at Noon	651	764	808	832	851	849	840	830	804	740	646	588	
	Edh at Noon	59	77	102	123	125	125	122	109	91	72	56	50	
All-Sky Solar Radiation	RadAvg	0.69	1.38	2.45	3.78	4.72	4.93	4.79	3.94	2.92	1.68	0.86	0.56	
	RadStd	0.07	0.16	0.29	0.44	0.31	0.47	0.51	0.27	0.27	0.14	0.10	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page